



Lecture 1.3: Per-Unit System

EEE 321: Power System I

tareq.aziz@bracu.ac.bd

Presentation Outline

Representation of Power System

- ❑ Per-unit system
- ❑ Examples and practices

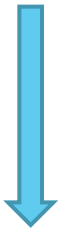
Necessity of Per Unit System

- ❑ In power systems large amounts of power being transmitted in the range of kilowatts to megawatts, at different voltage levels.
- ❑ As a result, in analysis, it is useful to scale, or normalize quantities with large physical values and this is commonly called per unit system in power system analysis.

Definition of Per Unit System

The per unit value of any quantity is defined as the ratio of the actual value of the quantity to its base value expressed as a decimal.

Quantity in per unit = Actual value of the quantity / Base value of the quantity



$$S_{pu} = \frac{S}{S_B}, \quad V_{pu} = \frac{V}{V_B}, \quad I_{pu} = \frac{I}{I_B} \quad \text{and} \quad Z_{pu} = \frac{Z}{Z_B}$$

where, S (apparent power), V (voltage), I (current) and Z (impedance) are phasor or complex quantities and denominators (i.e., S_B , V_B , I_B and Z_B) are always real numbers. To completely define a per-unit system, minimum four base quantities are required.

Benefits of Per Unit System

- ❑ For circuits connected by the transformers, per unit system is particularly suitable. Calculations are simplified because quantities expressed as per-unit do not change when they are referred from one side of a transformer to the other. This can be a pronounced advantage in power system analysis where large numbers of transformers may be encountered. For a transformer, when impedance in ohm is specified, it must be clearly mentioned whether it is with respect to primary or secondary. The per-unit impedance of the transformer, once expressed on proper base, is the same referred to either side. The way in which the three phase transformers are connected does not affect the per-unit impedances although the transformer connection does determine the relation between the voltage bases on the two sides of the transformer.
- ❑ Moreover, similar types of apparatus will have the impedances lying within a narrow numerical range when expressed as a per unit fraction of the equipment rating, even if the unit size varies widely. Thus, representations of elements in the system with per unit values become more uniform. Different types of quantities are labeled with the same symbol (pu); it should be clear from context whether the quantity is a voltage, current, etc.
- ❑ Conversion of per-unit quantities to volts, ohms, or amperes requires a knowledge of the base that the per-unit quantities were referenced to. The per-unit system is used in power flow, short circuit evaluation, etc.
- ❑ Manufacturers usually specify the impedance of a piece of apparatus in percent or per-unit on the base of the name plate rating.

Explanation of Per Unit System

For instance, if a base voltage of 120 kV is chosen, voltages of 108, 120 and 126 kV become 0.90, 1.00, and 1.05 per unit. In the power systems analysis, a per-unit system is the expression of system quantities as fractions of a defined base unit quantity.

- ❑ All base values are only magnitude. They are not associated with any angle.
- ❑ The per unit values, however, are phasors.
- ❑ The phase angles of the currents and voltages and the power factor of the circuit are not affected by the conversion to per unit values.
- ❑ In general, the per unit value is the ratio of the actual value and the base value of the same quantity.
- ❑ The per unit system values can also be expressed as per cent values.
- ❑ Usually, the nominal apparent power (S) and nominal voltage (V) are taken as the base values for power (S_{base}) and voltage (V_{base}).
- ❑ The base values for the current (I_{base}) and impedance (Z_{base}) can be calculated based on the first two base values.

Per Unit for 1 Φ System

For a power system, practical choice of base values are:

$$\text{Base megavoltamperes} = (\text{MVA})_B$$

or

$$\text{Base kilovoltamperes} = (\text{kVA})_B$$

$$\text{Base kilovolts} = (\text{kV})_B$$

$$\text{Base current } I_B = \frac{1,000 \times (\text{MVA})_B}{(\text{kV})_B} = \frac{(\text{kVA})_B}{(\text{kV})_B} \text{ A}$$

$$\begin{aligned} \text{Base impedance } Z_B &= \frac{1,000 \times (\text{kV})_B}{I_B} \\ &= \frac{(\text{kV})_B^2}{(\text{MVA})_B} = \frac{1,000 \times (\text{kV})_B^2}{(\text{kVA})_B} \text{ ohms} \end{aligned}$$

Per Unit for 3Φ System

In a three-phase system rather than obtaining the per unit values using per phase base quantities, the per unit values can be obtained directly by using three-phase base quantities. Let

Three-phase base megavoltamperes = $(\text{MVA})_B$

Line-to-line base kilovolts = $(\text{kV})_B$

Assuming star connection (equivalent star can always be found),

$$\text{Base current } I_B = \frac{1,000 \times (\text{MVA})_B}{\sqrt{3} (\text{kV})_B} \text{ A}$$

$$\begin{aligned} \text{Base impedance } Z_B &= \frac{1,000 \times (\text{kV})_B}{\sqrt{3} I_B} \\ &= \frac{(\text{kV})_B^2}{(\text{MVA})_B} = \frac{1,000 \times (\text{kV})_B^2}{(\text{kVA})_B} \text{ ohms} \end{aligned}$$

Example#1

- Given base kVA for 3 phase systems is 30 000kVA and voltage base line to line 120kV. Find:

$$S_{B,1\phi}$$

$$V_{pu}$$

$$V_{B,LN}$$

$$V_{LN}$$

- Given actual line to line voltage is 108kV.

Example#1

Solution

$$\text{Base kVA}_{3\phi} = 30,000 \text{ kVA}$$

and

$$\text{Base kV}_{LL} = 120 \text{ kV}$$

therefore $\text{Base kVA}_{1\phi} = 30,000 / 3 = 10,000 \text{ kVA}$

and $\text{Base kV}_{LN} = 120 / \sqrt{3} = 69.2 \text{ kV}$

For actual line-to-line voltage 108 kV, the line-to-neutral voltage, V_{LN} is $108 / \sqrt{3} = 62.3 \text{ kV}$

Per-unit voltage $= 108/120 (3\phi) \text{ OR}$
 $= 62.3/69.2 (1\phi)$
 $= 0.9$

$$\begin{array}{l} S_{B,1\phi} \\ V_{pu} \\ V_{B,LN} \\ V_{LN} \end{array}$$

Change of Base

- ❑ Some times the per unit impedance of a component of a system is expressed on a base other than the one selected as base for the part of the system in which the component is located.
- ❑ Since all impedances in any part of a system must be expressed on the same impedance base when making computations, it is necessary to have a means of converting per unit impedances from one base to another.

$$Z_{actual} = Z_{new}^{pu} Z_{B,new} = Z_{old}^{pu} Z_{B,old}$$

$$\Rightarrow Z_{new}^{pu} = Z_{old}^{pu} \frac{Z_{B,old}}{Z_{B,new}}$$

$$\text{But we always have: } Z_B = \frac{V_B^2}{S_B}$$

$$Z_{new}^{pu} = Z_{old}^{pu} \frac{S_{B,new}}{S_{B,old}} \left(\frac{V_{B,old}}{V_{B,new}} \right)^2$$

Conversion of per-unit impedance from one base to another base.

Example#2

The reactance of a generator designated X'' is given as 0.25 per unit based on the generator's nameplate rating of 18 kV, 500 MVA. The base for calculations is 20kV, 100 MVA. Find X'' on the new base.

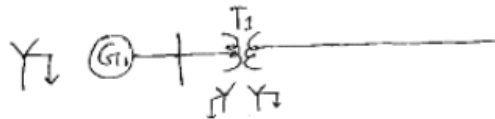
$$Z_{pu}^{new} = Z_{pu}^{old} \left(\frac{S_B^{new}}{S_B^{old}} \right) \left(\frac{V_B^{old}}{V_B^{new}} \right)^2$$

$$Z_{pu}^{old} = 0.25 \quad V_B^{old} = 18 \text{ kV} \quad S_B^{old} = 500 \text{ MVA}$$

$$V_B^{new} = 20 \text{ kV} \quad S_B^{new} = 100 \text{ MVA}$$

$$Z_{pu}^{new} = X'' = 0.25 \left(\frac{100}{500} \right) \left(\frac{18}{20} \right)^2 = 0.0405 \text{ per unit}$$

Practice#1



120 MVA
19.5 kV
 $X=1.5$ pu

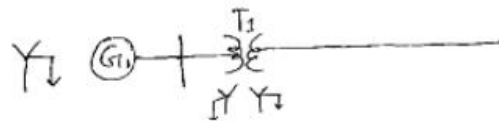
150 MVA
18/230 kV
 $X=0.1$ pu

$$Z_{pu}^{new} = Z_{pu}^{old} \left(\frac{S_B^{new}}{S_B^{old}} \right) \left(\frac{V_B^{old}}{V_B^{new}} \right)^2$$

Find the per unit reactance of components assuming base 100 MVA, 230 kV for the transmission line

Component	V_b old	V_b new	S_b old	S_b new	X_{pu} new
G					
Transformer					

Practice#1



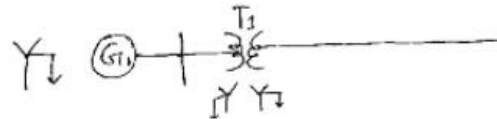
120 MVA
19.5 kV
X=1.5 pu

150 MVA
18/230 kV
X=0.1 pu

$$Z_{pu}^{new} = Z_{pu}^{old} \left(\frac{S_B^{new}}{S_B^{old}} \right) \left(\frac{V_B^{old}}{V_B^{new}} \right)^2$$

Find the per unit reactance of components assuming base 100 MVA, 230 kV for the transmission line

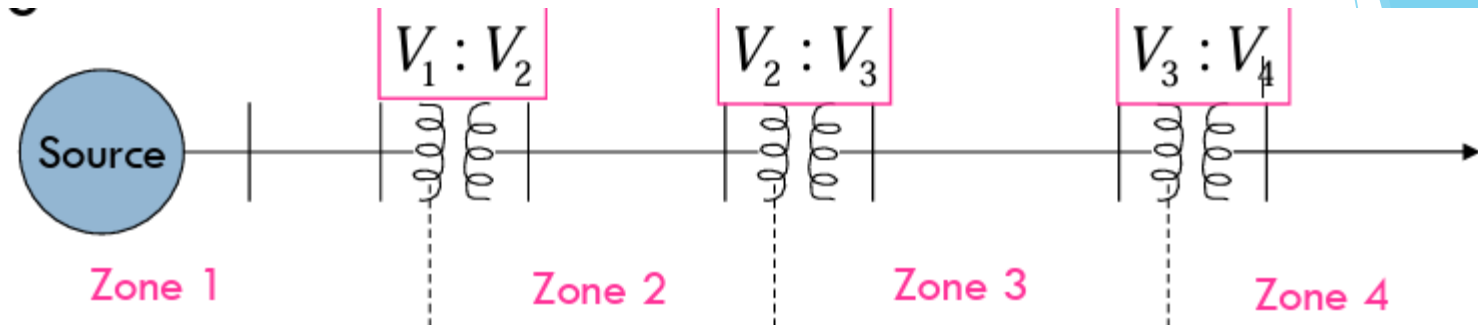
Solution:



V_b old	19.5	18/230	230
V_b new	18	18/230	230

Component	V_b old	V_b new	S_b old	S_b new	X_{pu} new
G	19.5	18	120	100	$1.5 \times \frac{19.5^2}{18^2} \times \frac{100}{120} = 1.467 \text{ pu}$
Transformer	18	18	150	100	$0.10 \times \frac{18^2}{18^2} \times \frac{100}{150} = 0.067 \text{ pu}$

Per Unit Impedance (Reactance) Diagram of a Power System



1. Choose an appropriate common MVA (or kVA) base for the system.
2. Consider the system to be divided into a number of sections by the transformers. Choose an appropriate kV base in one of the sections. Calculate kV bases of other sections in the ratio of transformation.
3. Calculate per unit values of voltages and impedances in each section and connect them up as per the topology of the one-line diagram. The result is the single-phase per unit impedance diagram.

Example#3 A 100 MVA, 33 kV, three phase generator has a reactance of 15%. The generator is connected to the motors through a transmission line and transformers as shown in Fig. 5.14. Motors have rated inputs of 40 MVA, 30 MVA and 20 MVA at 30 kV with 20% reactance-each. Draw the per-unit circuit diagram.

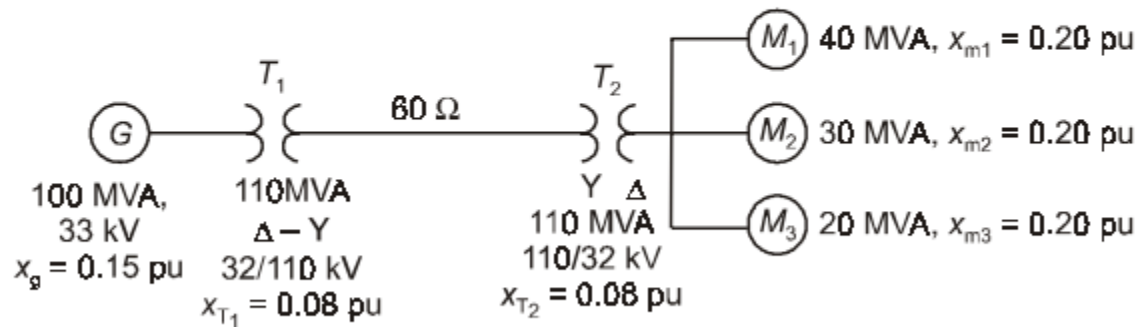


Fig. 5.14: Single-line diagram.

Solution: Assuming,

$(\text{MVA})_B = 100$ and $(\text{KV})_B = 33$ in the generator circuit.

$$\therefore x_g = 0.15 \text{ pu}$$

$$(\text{KV})_{B, \text{ line}} = 33 \times \frac{110}{32} = 113.43 \text{ kV}$$

In the motor circuit

$$(\text{KV})_{B, \text{ motor}} = 113.43 \times \frac{32}{110} = 33 \text{ kV.}$$

$$X_{T1} = 0.08 \times \frac{100}{110} \times \left(\frac{32}{33}\right)^2 = 0.0683 \text{ pu}$$

$$X_{T2} = 0.0683 \text{ pu}$$

Example#3 A 100 MVA, 33 kV, three phase generator has a reactance of 15%. The generator is connected to the motors through a transmission line and transformers as shown in Fig. 5.14. Motors have rated inputs of 40 MVA, 30 MVA and 20 MVA at 30 kV with 20% reactance-each. Draw the per-unit circuit diagram.

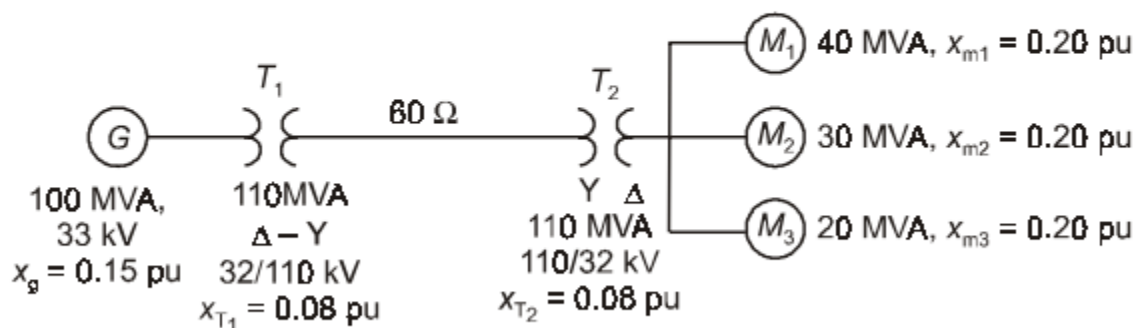


Fig. 5.14: Single-line diagram.

$$Z_{B, \text{line}} = \frac{(113.43)^2}{100} = 128.66 \, \Omega$$

$$x_{\text{line}}(\text{pu}) = \frac{60}{128.66} = 0.466 \, \text{pu}$$

Figure 5.15 shows the per-unit reactance diagram.

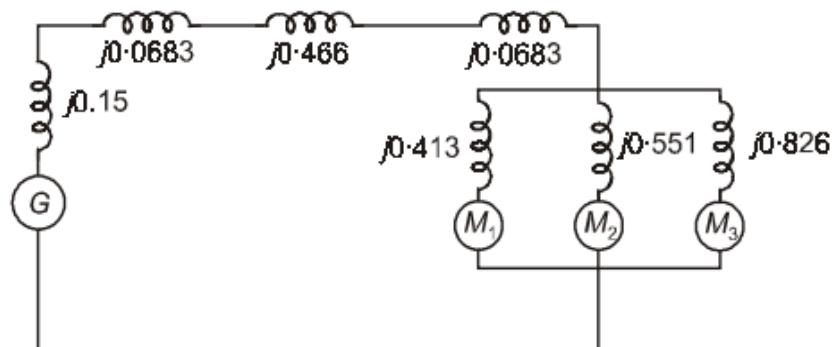


Fig. 5.15: Per-unit reactance diagram.

$$x_{\text{motor-1}}(\text{pu}) = 0.20 \times \frac{100}{40} \times \left(\frac{30}{33}\right)^2 = 0.413 \, \text{pu}$$

$$x_{\text{motor-2}}(\text{pu}) = 0.20 \times \frac{100}{30} \times \left(\frac{30}{33}\right)^2 = 0.551 \, \text{pu}$$

$$x_{\text{motor-3}}(\text{pu}) = 0.20 \times \frac{100}{20} \times \left(\frac{30}{33}\right)^2 = 0.826 \, \text{pu}.$$

Per Unit Impedance (Reactance) Diagram of a Power System

Example#4 Figure 5.10 shows single line diagram of a single-phase circuit. Using the base values of 3 kVA and 230 volts, draw the per-unit circuit diagram and determine the per-unit impedances and the per-unit source voltage. Also calculate the load current both in per unit and in Amperes.

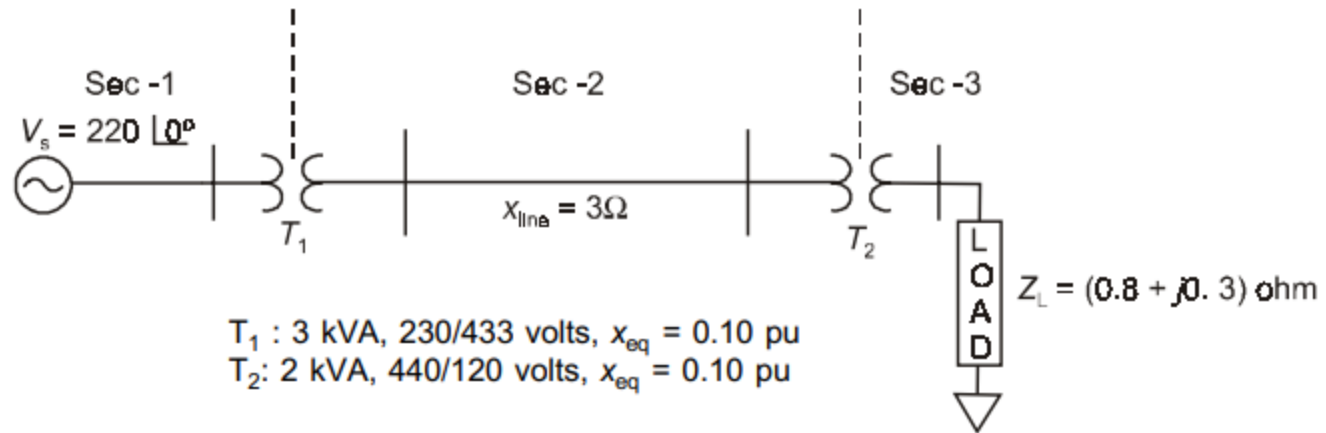


Fig. 5.10: Single-phase circuit.

Solution: First base values in each section have to be obtained.

Base MVA = $\frac{3}{1000} = 0.003$ and this base value will remain same for the entire network.

$\therefore$ (MVA)_B = 0.003

Also, $V_{B1} = 230$ volts = 0.23 kV, as specified in Section-1.

When moving across a transformer, the voltage base is changed in proportion to the transformer voltage ratings. Therefore,

Per Unit Impedance (Reactance) Diagram of a Power System

Example#4 Figure 5.10 shows single line diagram of a single-phase circuit. Using the base values of 3 kVA and 230 volts, draw the per-unit circuit diagram and determine the per-unit impedances and the per-unit source voltage. Also calculate the load current both in per unit and in Amperes.

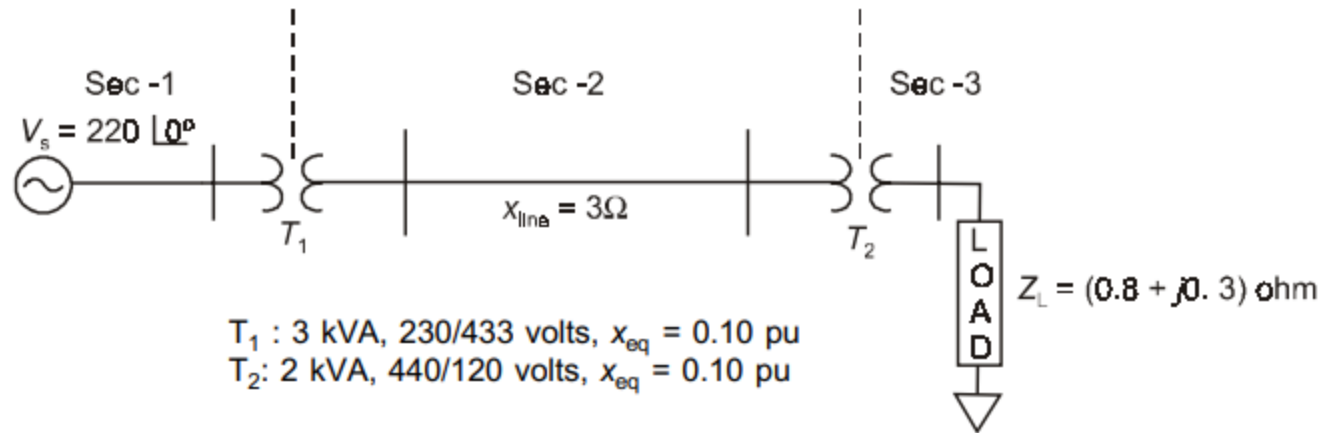


Fig. 5.10: Single-phase circuit.

$$V_{B2} = \left(\frac{433}{230} \right) \times 230 = 433 \text{ volt} = 0.433 \text{ kV}$$

$$V_{B3} = \left(\frac{120}{440} \right) \times 433 \text{ volts} = 118.09 \text{ volts} = 0.11809 \text{ kV.}$$

$$Z_{B1} = \frac{(V_{B1})^2}{(\text{MVA})_B} = \frac{(0.23)^2}{0.003} = 17.63 \text{ ohm}$$

$$Z_{B2} = \frac{(V_{B2})^2}{(\text{MVA})_B} = \frac{(0.433)^2}{0.003} = 62.5 \text{ ohm}$$

$$Z_{B3} = \frac{(V_{B3})^2}{(\text{MVA})_B} = \frac{(0.11809)^2}{0.003} = 4.64 \Omega$$

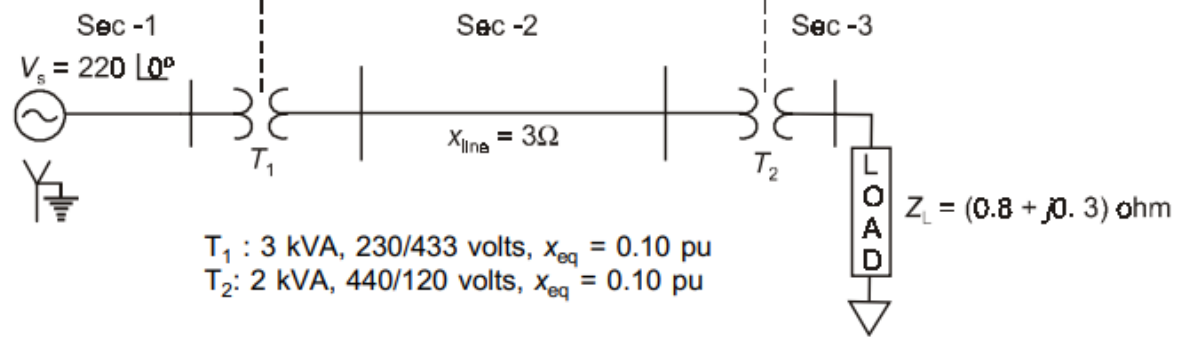


Fig. 5.10: Single-phase circuit.

Example#4

Base current :

$$I_{B3} = \frac{(MVA)_B}{(V_{B3})} = \frac{0.003}{0.11809} \text{ kA} = 25.4 \text{ Amp}$$

Given that $x_{1, \text{old}} = x_{eq} = 0.10$ pu

$\therefore x_{1, \text{new}} = 0.10$ pu

Therefore, for transformer- T_1 , no change in per-unit value of leakage reactance

For transformer, T_2 ,

$\therefore x_{2, \text{new}} = 0.1548$ pu

$$x_{\text{line}} (\text{pu}) = \frac{x_{\text{line}} (\text{ohm})}{Z_{B2}} = \frac{3}{62.5} = 0.048 \text{ pu}$$

$$Z_{\text{new}}^{\text{pu}} = Z_{\text{old}}^{\text{pu}} \frac{S_{B, \text{new}}}{S_{B, \text{old}}} \left(\frac{V_{B, \text{old}}}{V_{B, \text{new}}} \right)^2$$

Example#4

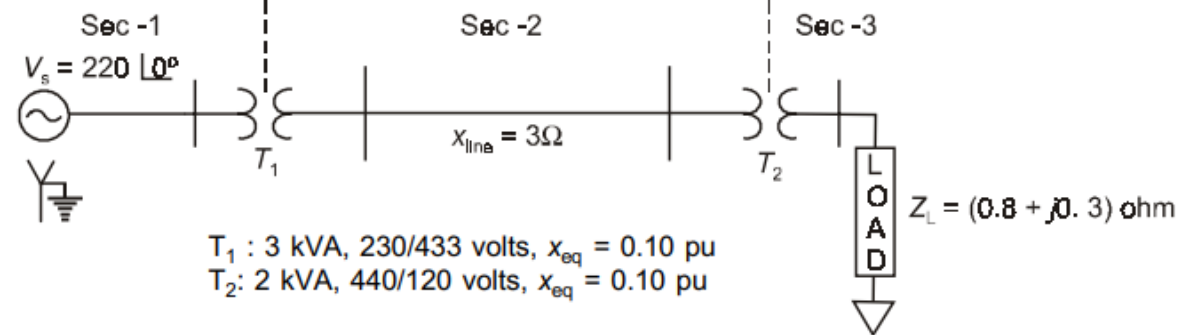


Fig. 5.10: Single-phase circuit.

$$Z_L(\text{pu}) = \frac{Z_L(\text{ohm})}{Z_{B3}} = \frac{(0.8 + j0.3)}{4.64}$$

$$= (0.1724 + j0.0646) \text{ pu.}$$

Per-unit circuit is shown in Fig. 5.11.

$$I_L(\text{pu}) = \frac{V_s(\text{pu})}{Z_T(\text{pu})}$$

$$Z_T(\text{pu}) = j0.10 + j0.048 + j0.1548$$

$$+ 0.1724 + j0.0646$$

$$= 0.4058 \angle 64.86^\circ$$

$$\therefore I_L(\text{pu}) = \frac{0.956 \angle 0^\circ}{0.4058 \angle 64.86^\circ} = 2.355 \angle -64.86^\circ \text{ pu}$$

$$I_L(\text{Amp}) = I_L(\text{pu}) \times I_{B3} = 2.355 \angle -64.86^\circ \times 25.4$$

$$= 59.83 \angle -64.86^\circ \text{ Amp.}$$

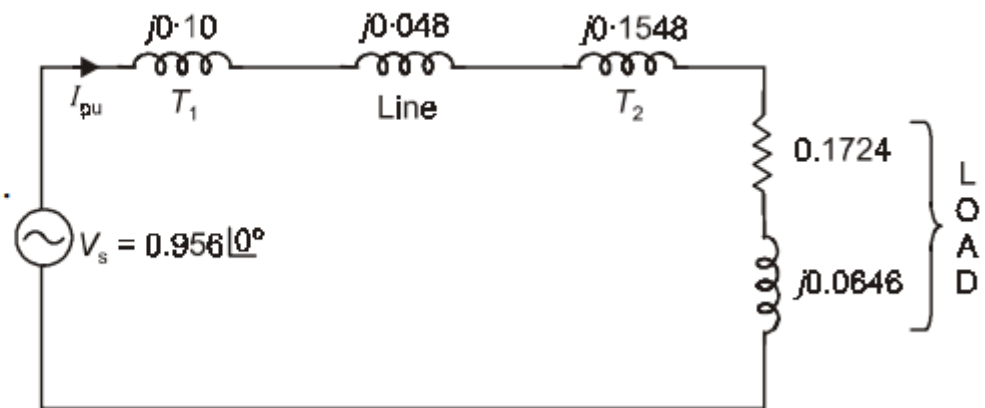


Fig. 5.11: Per-unit circuit.

Practice#2

The one-line diagram of a three-phase power system is shown in Figure 3.29. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow:

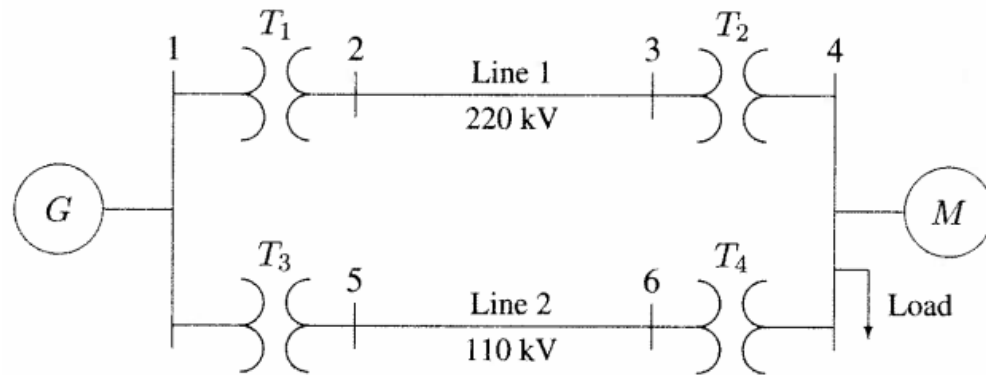


FIGURE 3.29

One-line diagram for Example 3.7.

G :	90 MVA	22 kV	$X = 18 \%$
T_1 :	50 MVA	22/220 kV	$X = 10 \%$
T_2 :	40 MVA	220/11 kV	$X = 6.0 \%$
T_3 :	40 MVA	22/110 kV	$X = 6.4 \%$
T_4 :	40 MVA	110/11 kV	$X = 8.0 \%$
M :	66.5 MVA	10.45 kV	$X = 18.5 \%$

The three-phase load at bus 4 absorbs 57 MVA, 0.6 power factor lagging at 10.45 kV. Line 1 and line 2 have reactances of 48.4 and 65.43 Ω , respectively.

Practice#2

The one-line diagram of a three-phase power system is shown in Figure 3.29. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow:

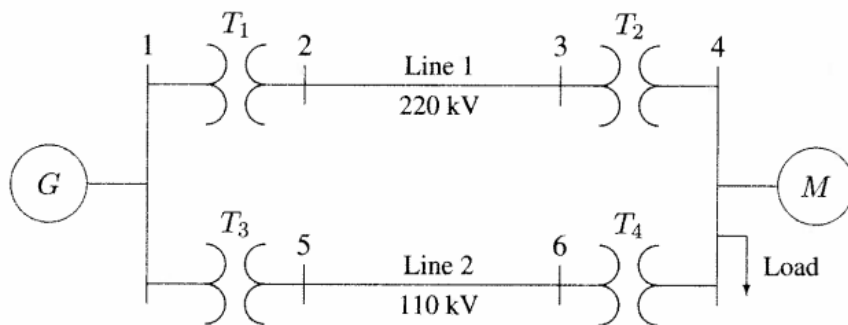


FIGURE 3.29
One-line diagram for Example 3.7.

G :	90 MVA	22 kV	$X = 18 \%$
T_1 :	50 MVA	22/220 kV	$X = 10 \%$
T_2 :	40 MVA	220/11 kV	$X = 6.0 \%$
T_3 :	40 MVA	22/110 kV	$X = 6.4 \%$
T_4 :	40 MVA	110/11 kV	$X = 8.0 \%$
M :	66.5 MVA	10.45 kV	$X = 18.5 \%$

The three-phase load at bus 4 absorbs 57 MVA, 0.6 power factor lagging at 10.45 kV. Line 1 and line 2 have reactances of 48.4 and 65.43 Ω , respectively.

Solution:

First, the voltage bases must be determined for all sections of the network. The generator rated voltage is given as the base voltage at bus 1. This fixes the voltage bases for the remaining buses in accordance to the transformer turns ratios. The base voltage V_{B1} on the LV side of T_1 is 22 kV. Hence the base on its HV side is

$$V_{B2} = 22 \left(\frac{220}{22} \right) = 220 \text{ kV}$$

This fixes the base on the HV side of T_2 at $V_{B3} = 220$ kV, and on its LV side at

$$V_{B4} = 220 \left(\frac{11}{220} \right) = 11 \text{ kV}$$

Similarly, the voltage base at buses 5 and 6 are

$$V_{B5} = V_{B6} = 22 \left(\frac{110}{22} \right) = 110 \text{ kV}$$

Practice#2

The one-line diagram of a three-phase power system is shown in Figure 3.29. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow:

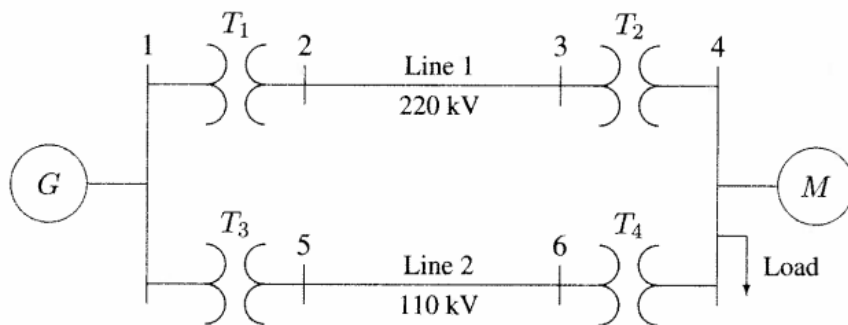


FIGURE 3.29
One-line diagram for Example 3.7.

G :	90 MVA	22 kV	$X = 18\%$
T_1 :	50 MVA	22/220 kV	$X = 10\%$
T_2 :	40 MVA	220/11 kV	$X = 6.0\%$
T_3 :	40 MVA	22/110 kV	$X = 6.4\%$
T_4 :	40 MVA	110/11 kV	$X = 8.0\%$
M :	66.5 MVA	10.45 kV	$X = 18.5\%$

The three-phase load at bus 4 absorbs 57 MVA, 0.6 power factor lagging at 10.45 kV. Line 1 and line 2 have reactances of 48.4 and 65.43 Ω , respectively.

Solution:

Since generator and transformer voltage bases are the same as their rated values, their per-unit reactances on a 100 MVA base, from (3.92) are

$$G: X = 0.18 \left(\frac{100}{90} \right) = 0.20 \text{ pu}$$

$$T_1: X = 0.10 \left(\frac{100}{50} \right) = 0.20 \text{ pu}$$

$$T_2: X = 0.06 \left(\frac{100}{40} \right) = 0.15 \text{ pu}$$

$$T_3: X = 0.064 \left(\frac{100}{40} \right) = 0.16 \text{ pu}$$

$$T_4: X = 0.08 \left(\frac{100}{40} \right) = 0.2 \text{ pu}$$

The motor reactance is expressed on its nameplate rating of 66.5 MVA and 10.45 kV. However, the base voltage at bus 4 for the motor is 11 kV. From (3.91) the motor reactance on a 100 MVA, 11-kV base is

$$M: X = 0.185 \left(\frac{100}{66.5} \right) \left(\frac{10.45}{11} \right)^2 = 0.25 \text{ pu}$$

Practice#2

The one-line diagram of a three-phase power system is shown in Figure 3.29. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow:

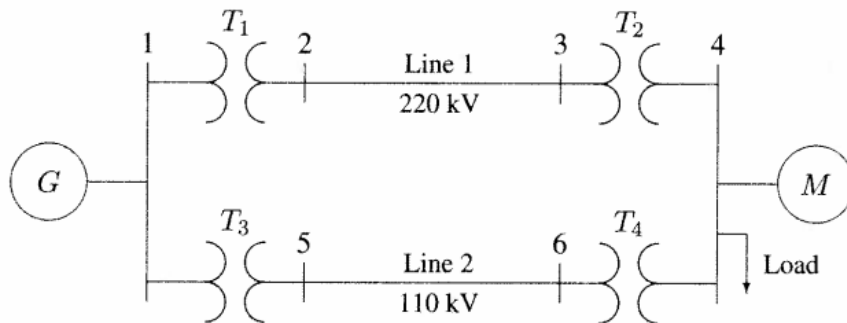


FIGURE 3.29
One-line diagram for Example 3.7.

G :	90 MVA	22 kV	$X = 18 \%$
T_1 :	50 MVA	22/220 kV	$X = 10 \%$
T_2 :	40 MVA	220/11 kV	$X = 6.0 \%$
T_3 :	40 MVA	22/110 kV	$X = 6.4 \%$
T_4 :	40 MVA	110/11 kV	$X = 8.0 \%$
M :	66.5 MVA	10.45 kV	$X = 18.5 \%$

The three-phase load at bus 4 absorbs 57 MVA, 0.6 power factor lagging at 10.45 kV. Line 1 and line 2 have reactances of 48.4 and 65.43 Ω , respectively.

Solution:

Line 1 and 2 per-unit reactances are

$$\text{Line 1: } X = \left(\frac{48.4}{484} \right) = 0.10 \text{ pu}$$

$$\text{Line 2: } X = \left(\frac{65.43}{121} \right) = 0.54 \text{ pu}$$

The load apparent power at 0.6 power factor lagging is given by

$$S_{L(3\phi)} = 57 \angle 53.13^\circ \text{ MVA}$$

Hence, the load impedance in ohms is

$$Z_L = \frac{(V_{L-L})^2}{S_{L(3\phi)}^*} = \frac{(10.45)^2}{57 \angle -53.13^\circ} = 1.1495 + j1.53267 \text{ } \Omega$$

Practice#2

The one-line diagram of a three-phase power system is shown in Figure 3.29. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow:

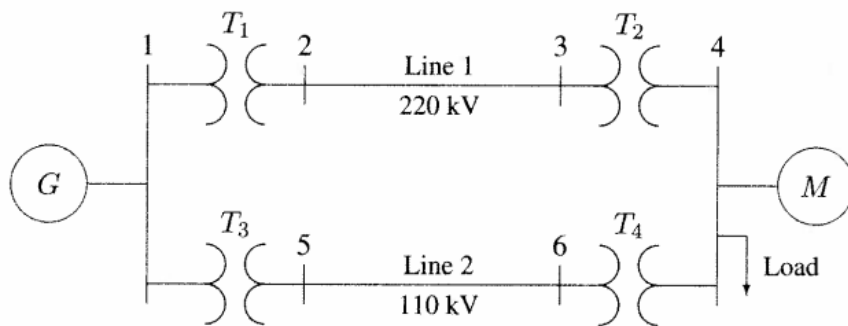


FIGURE 3.29
One-line diagram for Example 3.7.

G :	90 MVA	22 kV	$X = 18\%$
T_1 :	50 MVA	22/220 kV	$X = 10\%$
T_2 :	40 MVA	220/11 kV	$X = 6.0\%$
T_3 :	40 MVA	22/110 kV	$X = 6.4\%$
T_4 :	40 MVA	110/11 kV	$X = 8.0\%$
M :	66.5 MVA	10.45 kV	$X = 18.5\%$

The three-phase load at bus 4 absorbs 57 MVA, 0.6 power factor lagging at 11 kV. Line 1 and line 2 have reactances of 48.4 and 65.43 Ω , respectively.

Solution:

The base impedance for the load is

$$Z_{B4} = \frac{(11)^2}{100} = 1.21 \quad \Omega$$

Therefore, the load impedance in per-unit is

$$Z_{L(pu)} = \frac{1.1495 + j1.53267}{1.21} = 0.95 + j1.2667 \quad \text{pu}$$

The per-unit equivalent circuit is shown in Figure 3.30.

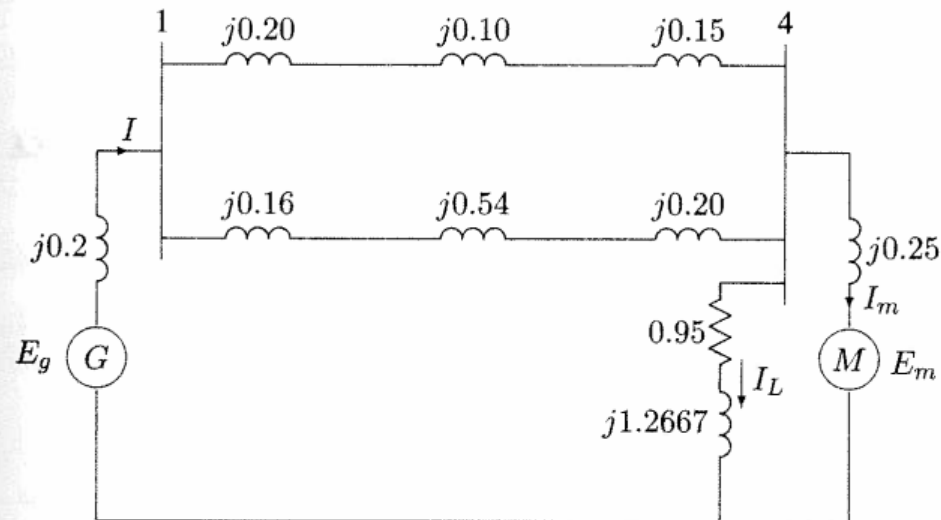


FIGURE 3.30
Per-unit impedance diagram for Example 3.7.

Reference: Power System Analysis, Hadi Saadat, Chapter 3.



QUESTIONS AND DISCUSSIONS

For further queries: tareq.aziz@bracu.ac.bd